

9. Orchestrating Serverless Functions with AWS Step Functions - simulation in Docker

Go to killercode.com select docker followed by build an image and perform the following tasks

Step 1: Create Folders and Basic Files

1. Create project folder and move into it

```
mkdir project  
cd project
```

2. Create subfolders for:

- input (where you upload the source image)
- output (where resized and watermarked images will be saved)
- resize (for first container: resize image)
- watermark (for second container: add watermark)

```
mkdir input output resize watermark
```

Now your folder tree looks like this:

```
/project
```

```
/input  
/output  
/resize  
/watermark
```

3. Create resize/app.py file

```
cd resize  
vi app.py
```

Type this Python code inside: (press i to insert)

```
from PIL import Image  
import os  
  
input_path = '/data/input/image.jpg'  
output_path = '/data/output/resized.jpg'  
  
print("Resizing image...")  
image = Image.open(input_path)  
image = image.resize((300, 300))  
image.save(output_path)  
print("Image resized and saved!")
```

Press Esc, and type :wq

4. Create resize/Dockerfile

```
vi Dockerfile
```

Type this inside:

(press i to insert)

```
FROM python:3.11-slim
```

```
RUN pip install pillow
```

```
WORKDIR /app
```

```
COPY app.py .
```

```
CMD ["python", "app.py"]
```

Save and exit. (Press ESC, :wq)



5. Now set up watermark/app.py

Go back to project root:

```
cd ../watermark
```

```
(or )
```

```
cd ..
```

```
cd watermark
```

Now create another python file here.

```
vi app.py
```

Type this python code:

```
from PIL import Image, ImageDraw, ImageFont
import os
```

```
input_path = '/data/output/resized.jpg'
```

```
output_path = '/data/output/watermarked.jpg'

print("Adding watermark...")
image = Image.open(input_path)
draw = ImageDraw.Draw(image)
font = ImageFont.load_default()
draw.text((50, 50), "Sample Watermark", fill=(0, 255, 128), font=font)
image.save(output_path)
print("Watermarked image saved!")
```

Save and exit. (ESC, :wq)

6. Create watermark/Dockerfile

You are inside the project/watermark ; Now create a Dockerfile here.

```
vi Dockerfile
```

Type this:

```
FROM python:3.11-slim
```

```
RUN pip install pillow
```

```
WORKDIR /app
```

```
COPY app.py .
```

```
CMD ["python", "app.py"]
```

Save and exit.

 **Now your full structure looks like this:**

```
/project
  /input/
  /output/
  /resize/
    app.py
    Dockerfile
  /watermark/
    app.py
    Dockerfile
```

Now Build Containers:

Build Containers

In your Play-with-Docker terminal:

```
cd /project
```

```
{
```

OR just type:

```
cd ..
```

This will take you to /projects }

```
# Build resize container
```

```
docker build -t resize-image ./resize
```

```
# Build watermark container
```

```
docker build -t watermark-image ./watermark
```

Now add the image: Type the following command:

```
wget -O input/image.jpg https://picsum.photos/600/400
```

Run Containers in Sequence (simulate Step Functions)

```
# Resize the image
```

```
docker run --rm -v $(pwd)/input:/data/input -v  
$(pwd)/output:/data/output resize-image
```

```
# Watermark the resized image
```

```
docker run --rm -v $(pwd)/input:/data/input -v  
$(pwd)/output:/data/output watermark-image
```

Now run the following Command:

```
python3 -m http.server 8080
```

Now click on open port and type 8080

You will get the name of different folders created under the directory named project.

Click on the output folder and you will see the links to open resized-image and watermarked-image. Click on them and see the results. That's it.

← → ↻ ⚠ Not secure ip172-18-0-73-d0gti0291nsg008pkelg-8080.direct.labs.play-with-docker.com

Directory listing for /

- [input/](#)
 - [output/](#)
 - [resize/](#)
 - [watermark/](#)
-

After selecting the input directory:

← → ↻ ⚠ Not secure ip172-18-0-73-d0gti0291nsg008pkelg-8080.direct.labs.play-with-docker.com/input/

Directory listing for /input/

- [image.jpg](#)
-

Once you Click on the image.jpg you will get the following image:



Once you Click on Output directory you will the following:

← → ↻ ⚠ Not secure ip172-18-0-73-d0gti0291nsg008pkelg-8080.direct.labs.play-with-docker.com/output/

Directory listing for /output/

- [resized.jpg](#)
- [watermarked.jpg](#)

Once you click on the resize.jpg the resized image will appear:



Once you Click on the [watermarked.jpg](#), you will get the following watermarked image:

